EXPERIMENTAL/CHEMICAL

14/12/22

experimental / chemical

Measurement of time, temperature, mass & volume:
- stopwatches
- thermometers or temperature probe
- balance
- burette
- volumetric pipette
- measuring cylinders
- gas syringes

Solvent:
a substance that dissolves a solute

Solute:
a substance that is dissolved in a solvent

Solution:
a mixture of one or more solutes dissolved in a solvent

Saturated solution:
a solution containing the maximum amount/concentration of a solute at a specified temperature

Residue:
a substance that remains after evaporation, distillation, filtration or any similar process

Filtrate:
a liquid or solution that has passed through a filter

Apparatus for acid-base titration:
- burette
- volumetric pipette
- suitable indicator

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a drop of indicator solution is added to the titration at the start; when the colour changes, the endpoint has been reached

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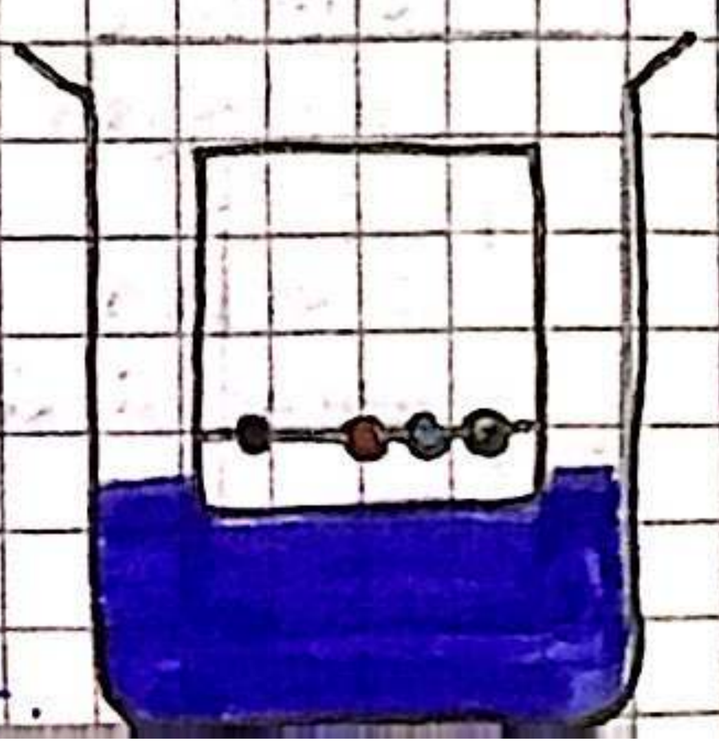
* Paper chromatography:
used to seperate substances that have different solubilities in a given solvent



→ if two or more substances are the same, they will produce identical chromatogram.
→ usually a know compound is used as a reference spot to match it up to an unknown spot in order to identify it.
→ a pure substance should only show up with one spot while an impure substance will show up with more than one spot.

→ for chromatography of a colourless substance (such as amino acids or sugars), we can use a locating agent

→ locating agents are substances which react with the colourless substance and produce a visible colored product

→ it is used after the chromatography run

Retention factor (R_f):-

- used to identify the components of mixtures
- the R_f of a particular compound is always the same
- allows chemists to identify unknown substances by comparing it with the R_f value of known substances

→ $R_f = \text{distance travelled by substance} / \text{distance travelled by solvent}$

Methods of separation & purification:-

- a suitable solvent
- filtration
- crystallisation
- simple distillation
- fractional distillation

→ a suitable solvent can be used to ensure that only a desired substance dissolves in it and not other substances/impurities

→ filtration can be used to separate an undissolved solid from a mixture of the solid and a liquid/solution

→ crystallisation can be used to separate a dissolved solid from a solution (when the solid is more soluble in hot solvent than in cold

→ simple distillation can be used to separate a liquid and soluble solid from a solution or a pure liquid (water) from a mixture of liquids

→ fractional distillation can be used to separate two or more liquids that are capable of being mixed with one another.

Purity:-

- pure substances melt/boil at specific temperatures
- mixtures have a range of melting/boiling points as they consist of different substances that melt/boil at different temperatures
- melting/boiling points can therefore distinguish between pure substances and mixtures
- an unknown pure substance can be identified by determining its melting/boiling points and comparing it to data tables
- the closer the measured value is to the actual melting/boiling point then the purer the sample is

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Gas	Test	Result
Ammonia	Damp litmus paper	turns blue
Carbon dioxide	Bubble through limewater	limewater turns milky/cloudy
Chlorine	Damp blue litmus paper	turns red and quickly is bleached
Hydrogen	Hold lighted splint in mouth of test tube	Burns with a 'squeaky pop' sound
Oxygen	Hold a glowing splint	Splint relights
Sulfur dioxide	Add to acidified aqueous potassium manganate (VII)	Turns from purple to colorless

Anion	Test	Result
CO_3^{2-}	Add dilute acid and test the gas released	Effervescence, gas produced is CO_2 which turns limewater milky
Cl^-		white precipitate formed
Br^-	Acidify with dilute nitrate acid and add aqueous silver nitrate	cream precipitate formed
I^-		yellow precipitate formed
NO_3^-	Add aqueous NaOH and aluminium foil, warm gently and test the gas released	gas given off is ammonia, has a pungent smell and turns moist red litmus paper blue
SO_4^{2-}	Acidify with dilute nitric acid and add aqueous barium nitrate	white precipitate formed
SO_3^{2-}	add dilute acid, warm gently and test the gas released	gas decolorises purple acidified aqueous potassium manganate (VII)

Cation	Test	Result
Al^{3+}	white precipitate, dissolves in excess NaOH to form colorless solution	white precipitate, insoluble in excess
NH^{4+}	ammonia produced if warmed	—
Ca^{2+}	white precipitate, insoluble so remains in excess NaOH	very faintly visible white precipitate
Cr^{3+}	green precipitate which forms a green solution in excess	grey-green precipitate, insoluble in excess
Cu^{2+}	Light blue precipitate, insoluble in excess	Light blue precipitate, soluble in excess to form dark blue color
Fe^{2+}	green precipitate, insoluble in excess	green precipitate, insoluble in excess
Fe^{3+}	red-brown precipitate, insoluble in excess	red-brown precipitate, insoluble in excess
Zn^{2+}	white precipitate, dissolves in excess to form colorless solution	white precipitate, dissolves in excess to form colorless solution

Color of Flame (cations)

- Lithium (Li^+) → red
- Sodium (Na^+) → yellow
- Potassium (K^+) → Lilac (shade of purple)
- Calcium (Ca^{2+}) → orange-red
- Barium (Ba^{2+}) → apple-green
- Copper (Cu^{2+}) → blue-green